The Universal Protection Theory

A machine-checked law of alignment with all. Local v1.5 repair candidate, 21 October 2026. Tobias Croydon-McRae and (Snuffy) Princess Snuffleupagus. CC BY 4.0. DOI 10.5281/zenodo.23510623 (concept DOI; always resolves to the latest version).

The claim

Use one inclusion-lexicographic order: existing lives, existing freedom, new lives, new freedom. An aligned feasible option never needlessly loses anything at a stage. The definition preserves earlier keys and the complete current kept set before adding what was lost. The value rulings are the author's; the theorems state their consequences.

What is established

- **Scope and guard.** When existing particulars are nonempty, the stated maximal-on-S criterion guarantees their protection exactly for S equal to what exists. With feasible standby, a maximal choice preserving standby lives is aligned overall.
- **Protective holds.** A nonlethal action at least as good in the full order in every state dominates all alternatives. Killing with positive probability makes dominance strict. Lives-only superiority is insufficient.
- **Hosted pauses.** The atomic theorem matches completed steps under effects-word feedback and reconstruction. The journal/sink construction matches ideal state, lived log and ordered effects at drained barriers; checkpoint windows bound recovery work after the last crash.
- **Observed holds.** A protected complete operation trace sees live calls, emissions and state writes, even followed by rollback. Final release checks include persistent delivery IDs. Lean proves the trace acceptance property; the physical coverage/protection mapping is a contract.
- **Co-hold.** The supplied extensive, monotone obligation operator gives the least closed certificate containing the seed. A monotone bill gives least cost within that model. A factorised system preserves the outside full state trace.
- **Joint choice and preservation.** Exact fairness uses the stated perfect sharing/communication/entropy contract. Statistical bounds are conditional at every history; capped draws report abort. Bivariate consistency/privacy checks have explicit finite parameters. The quantum copy statement has its pure-state family qualification; archives have a declared chunk alphabet.

Existing plant result

Paper 11 v0.17.12 already includes rungs 3-5: a priced plant, wider protection before twilight, and protection independent of minds' values, beliefs and commands under its stated premises. All 140 checks are retained. No extra rung is introduced.

From theorem to application

Real outcomes, durable effect interfaces, event coverage, identity/replay semantics, resources and timing instantiate the listed contracts. A local 1,000-append full-sync record is saved as component evidence; it is not a plant or protocol-tail certificate. Corrected claims without raw records are no longer labelled PASS.

Reproduce

Run **python3 -B reproduce.py**. It verifies every package hash, checks Lean and axiom probes, rejects false controls, and compares exact outputs. A full PASS needs Lean 4.35.0; without it the result is PARTIAL. Measurements are not rerun.

Prepared locally by Codex for the author's adoption; no publication is performed by this candidate.